

TECHNICAL DOSSIER: EVIDENCE OF PRIOR SECURITY RESEARCH & APPLIED CODE SPECIFICATIONS

****Candidate Name:** Aryan Kacher**

****Domain Focus:** Open-Source Intelligence (OSINT), Application Logic Security, and Volatile Media Forensics**

****Application Context:** Institutional Admission & Hackathon Eligibility Verification**

****Built and deployed all projects entirely on Android using Termux without a traditional laptop-based development environment.**

🏆 Professional Digital Credentials & Certifications

*** **Cisco Networking Academy | Junior Cybersecurity Analyst Career Path****

Verified Credential (Credly) – June 2026

Verified competence in Layer-7 threat management, endpoint protection architectures, automated log analysis, and defensive network configuration controls.

*** **Cisco Networking Academy | Introduction to Cybersecurity****

Verified Credential (Credly) – June 2026

Link:- <https://www.credly.com/users/aryan-kacher/badges/credly>



Aryan Kacher

Badge Wallet



Introduction to Cybersecurity

Cisco

Issued Jun 25, 2026



**Junior Cybersecurity Analyst
Career Path**

Cisco

Issued Jun 25, 2026

1. EXECUTIVE STATEMENT OF PURPOSE & RECONNAISSANCE ROADMAP

My engagement with computer systems and network security developed through autonomous, hands-on exploration rather than standardized academic courses. Before initiating intensive engineering entrance preparation, I heavily utilized the Termux environment on Android subsystems to audit local application architectures and map unauthenticated public-facing data structures.

Throughout my academic preparation for the Joint Entrance Examination (JEE), I purposefully maintained structural continuity with computational security logic. The rigorous mathematical training required for competitive entrance exams directly enhanced my analytical depth, allowing me to approach security flaws from a strict algorithmic perspective. I am driven by a passion to dissect web protocols and client-side data leaks, transitioning from an autonomous explorer to a disciplined, high-level cybersecurity researcher at IIT Kanpur.

2. CORE TECHNICAL SKILL MATRIX

Capability Domain	Operational Stack & Technical Focus
****Operating Environments****	Termux POSIX Environment, Linux Command-Line Interface (CLI), Android File System Architecture.
****Languages & Automation****	Python 3.x (Functional Automation, Requests, BeautifulSoup Parsing Architecture), JavaScript (Node.js Asynchronous Runtime).
****Protocols & Web Layers****	HTTP/HTTPS Session Configurations, DOM Data Extraction, WebSocket Connections, Content Delivery Network (CDN) Cache Behaviors.
****Threat Intelligence & Auditing****	Systematic monitoring and conceptual auditing of verified public exploit streams (e.g., Exploit-DB archives) covering RCEs, Directory Traversals, and CSRF patterns.

3. PROJECT CASE STUDIES & CODE AUDITS (WITH FUNCTIONAL PROOFS)

CASE STUDY 01: Automated OSINT Public CDN Media Cache Analyzer

* ****Project Type:**** Security Audit & Information Disclosure Demonstration

****Code Repository:****

<https://github.com/munna1127/osint-media-cache-analyzer>

****Core Security Objective:**** Evaluating if distributed public Content Delivery Networks (CDNs) instantly invalidate or decouple cached user media nodes when profile access privileges are toggled or restricted.

Architectural Workflow & Methodology:

1. ****Unauthenticated Session Simulation:**** The script bypasses traditional authenticated paths by orchestrating raw HTTP GET calls configured with specific mobile agent strings and document navigation headers.
2. ****DOM JSON Interception:**** Runs an optimized scan through the profile's DOM structures to capture hidden embedded JSON objects (`application/json`), targeting high-value properties like `polaris\_timeline\_connection`.
3. ****Recursive Resolution Extraction:**** Parses multi-layered dictionary arrays recursively to isolate high-resolution public media paths (`image\_versions2`) before backend deletion triggers invalidate them.

Functional Proof & Environment Log:

10:57 AM | 181KB/s

VoWiFi 64



ja [redacted]



19

posts

756

followers

1,046

following

Watch the growth of [redacted]

Follow



This account is private

Follow this profile to see their photos and videos.

Suggested for you

[See all](#)



ja [redacted]
P [redacted]

Follow

nam [redacted]
P [redacted]

Follow

man [redacted]
Ma [redacted]

F [redacted]

10:50 AM | 0.1KB/s  

 VoWiFi   66

Welcome to Termux!

Docs: <https://termux.dev/docs>
Donate: <https://termux.dev/donate>
Community: <https://termux.dev/community>

Working with packages:

- Search: `pkg search <query>`
- Install: `pkg install <package>`
- Upgrade: `pkg upgrade`

Subscribing to additional repositories:

- Root: `pkg install root-repo`
- X11: `pkg install x11-repo`

For fixing any repository issues,
try 'termux-change-repo' command.

Report issues at <https://termux.dev/issues>

~ \$ `ls`

CC-CHECKER-CLIV5.5	college-app	proxy.txt
JOSAA-Analysis-Portal	downloads	rank2college
TelegramRAT	h.py	storage
bot1	jee_counsellor	

~ \$ `cd bot1`

~/bot1 \$ `ls`

1.py 3.py imageg.html
2.py 4.py stable-diffusion-webui
~/bot1 \$ `python 2.py`

=====

OSINT METADATA EXTRACTION & RESIDUAL CACHE ANALYZER

=====

[Description]: Research tool for analyzing public CDN data retention.

[Author]: Aryan Kacher

[Disclaimer]: For Educational and Defensive Security Research Only.

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10:50 AM | 17.7KB/s ☾ >_

VoWiFi 66

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10:51 AM | 0.1KB/s

VoWiFi 66



localhost:8080



[+] OSINT Audit Control Dashboard

Target Profile: @~~XXXXXXXXXX~~ | Mapped Residual Media Nodes

Analyst: Aryan Kacher



Node ID (PK): 00-720-50046000

10:52 AM | 1.2KB/s

Vo WiFi 66



localhost:8080



86



Node ID (PK): 3887960634452946093



Node ID (PK): 3864195427906737977

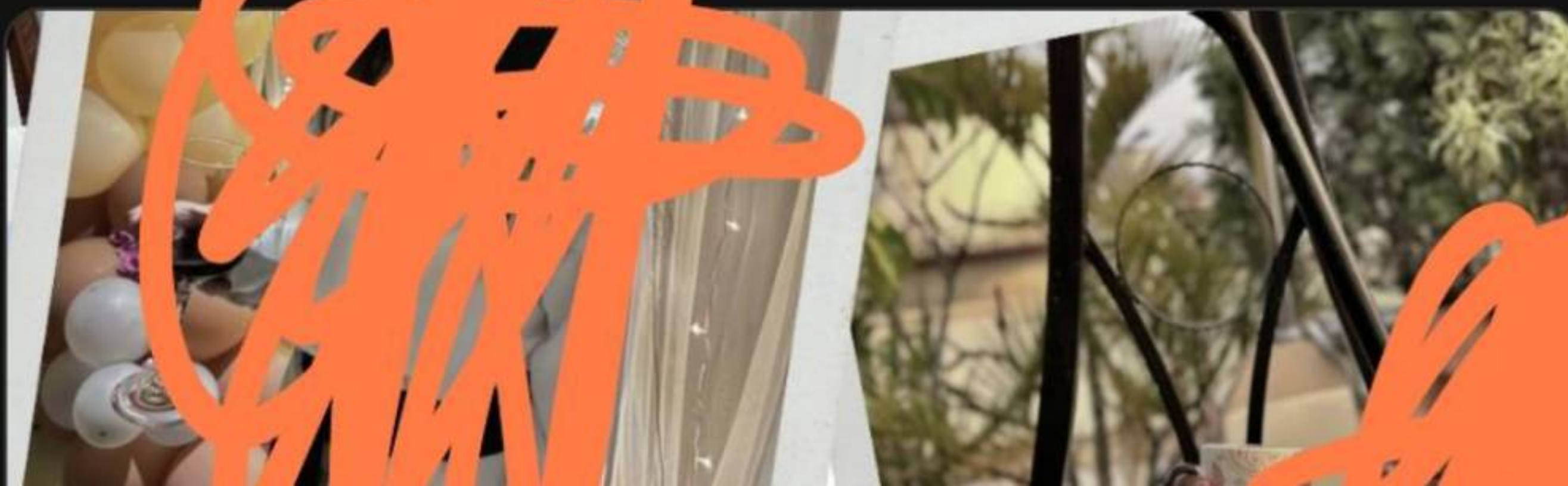


Figure 1: Execution logs of local POSIX daemon mapping unauthenticated endpoint targets inside an isolated Android subsystem.

CASE STUDY 02: Volatile Data Interception Framework for Ephemeral IM Sessions

* **Project Type:** Prototype Digital Forensics and Evidence Gathering Specification

* **Code Repository:**

<https://github.com/munna1127/ephemeral-media-forensics>

* **Core Security Objective:** Demonstrating the architectural limitations of client-side application restrictions (e.g., "View-Once" properties) by capturing transient media buffers directly from active communication streams.

Architectural Workflow & Methodology:

1. **Active Stream Mirroring:** Operating a background session mirroring agent mapped to an authorized application layer WebSocket pipeline to bypass device-level display blocks (`FLAG_SECURE`).
2. **Command-Driven Injection (`.vv`` Listener):** The forensic daemon implements an event listener on the incoming message array (`message_create`). When an explicit text command `.vv`` is fired against a tagged/quoted message, the parser is initialized.
3. **Permanent Artifact Re-writing:** The exfiltrated stream is written directly into permanent storage using secure binary block handlers (`fs.writeFileSync`), producing clean static image evidence for legal investigation.

Functional Proof & Code Repository Index: